Bitcoin Model Notes

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3.1 Model Setup

To concretize the idea in a tractable environment, consider a representative domestic investor who chooses consumption and portfolio holdings in discrete periods $t = 0, 1, 2, \ldots$. The investor can hold two risky assets: a domestic stock market claim and a global crypto asset (e.g., Bitcoin) that is priced in USD. At the end of each period, all wealth is marked to local currency (LCU). The investor receives exogenous labor income I_t in LCU.

Let C_t denote consumption and P_t^c the domestic price level (CPI, LCU). The gross inflation rate is

$$1 + \pi_{t+1} = \frac{P_{t+1}^c}{P_t^c}.$$

For the domestic stock market, let P_t^s be the stock price (LCU) and D_{t+1} the dividend (LCU) paid at t+1.

The gross *nominal* and *real* returns are

$$1 + R_{t+1}^s = \frac{P_{t+1}^s + D_{t+1}}{P_t^s}, \qquad 1 + R_{t+1}^{s,\mathrm{real}} = \frac{1 + R_{t+1}^s}{1 + \pi_{t+1}}.$$

For the cryptocurrency asset priced in USD, let $\Theta_{t+1}^\star$ be the USD return on the crypto asset (e.g., BTC) and define its log return $\theta_{t+1} \simeq \log(1 + \Theta_{t+1}^\star)$. Let S_t be the domestic exchange rate expressed as LCU per USD. The FX log return is

$$\Delta s_{t+1} = \log S_{t+1} - \log S_t,$$

where $\Delta s_{t+1} > 0$ means the dollar has appreciated against the LCU.

The investor chooses (i) the number of domestic shares n_t to carry from t to t+1 and (ii) the LCU amount A_t^k to allocate to the crypto leg at t. Investing A_t^k converts to a USD amount A_t^k / S_t at t, earns USD gross return $(1 + \Theta_{t+1}^\star)$ over $[t, t+1]$, and is repatriated at S_{t+1} into LCU at t+1.

Model setup. We consider a representative domestic investor who operates in discrete time $t = 0, 1, 2, \ldots$. The investor makes two key decisions each period: how much to consume (C_t), and how to allocate wealth across two risky assets. The first asset is a domestic stock market claim denominated in local currency (LCU). The second is a global cryptocurrency asset (e.g. Bitcoin) denominated in U.S. dollars (USD). At the end of each period, all positions are marked to LCU. The investor also receives an exogenous labor income stream I_t in LCU.

Inflation. Let P_t^c denote the domestic price level (CPI, in LCU). Inflation measures how the general price level changes between periods. The gross inflation rate is defined as

$$1 + \pi_{t+1} = \frac{P_{t+1}^c}{P_t^c}.$$

This formula says: take the price level tomorrow, divide it by today's price level, and you obtain the growth factor of prices. For example, if $P_t^c = 100$ and $P_{t+1}^c = 103$, then $1 + \pi_{t+1} = 103/100 = 1.03$, which means inflation was 3% over the period. This adjustment is important because it tells us how much of nominal returns simply reflect rising prices rather than real increases in purchasing power.

Domestic stocks. Let P_t^s be the stock price at time t (in LCU) and D_{t+1} the dividend paid at time $t+1$ (also in LCU). The gross nominal return is

$$1 + R_{t+1}^s = \frac{P_{t+1}^s + D_{t+1}}{P_t^s}.$$

The numerator combines both capital gains (price appreciation, $P_{t+1}^s - P_t^s$) and cash payouts (dividends, D_{t+1}). Dividing by P_t^s normalizes this by the initial price to compute the percentage growth factor. For example, suppose $P_t^s = 50$, $P_{t+1}^s = 52$, and $D_{t+1} = 1$. Then

$$1 + R_{t+1}^s = \frac{52 + 1}{50} = \frac{53}{50} = 1.06,$$

so the nominal return is 6%.

To adjust for inflation, we divide by the gross inflation rate:

$$1 + R_{t+1}^{s,\text{real}} = \frac{1 + R_{t+1}^s}{1 + \pi_{t+1}}.$$

Continuing the example, if inflation was 3%, then the real return is $1.06/1.03 \approx 1.029$, or about 2.9%. This shows that even though the stock nominally earned 6%, only 2.9% reflects an actual increase in real purchasing power after accounting for price inflation.

Cryptocurrency asset. For the cryptocurrency, denominated in USD, let Θ_{t+1}^* denote the gross return in USD terms. The log return is

$$\theta_{t+1} \simeq \log(1 + \Theta_{t+1}^*).$$

Logarithms are often used because they make compounded returns additive across time.

Since the investor is in LCU, the exchange rate matters. Let S_t be the domestic exchange rate, defined as LCU per USD. The gross FX return is the growth in S:

$$1 + R_{t+1}^{fx} = \frac{S_{t+1}}{S_t},$$

and its log return is

$$\Delta s_{t+1} = \log S_{t+1} - \log S_t.$$

If $\Delta s_{t+1} > 0$, then $S_{t+1} > S_t$, meaning more LCU are required per USD — the USD has appreciated relative to the LCU.

For example, suppose $S_t = 10$ LCU/USD and $S_{t+1} = 11$ LCU/USD. Then $\Delta s_{t+1} = \log(11) - \log(10) \approx 0.095$, or 9.5%. This appreciation means that holding any USD-denominated asset (like Bitcoin) results in an extra 9.5% gain purely from the exchange rate movement, independent of the crypto price change in USD.

Thus, for a domestic investor, the effective crypto return in LCU comes from two sources: the USD crypto return and the FX return. This dual dependence is why crypto acts differently from domestic stocks — it bundles asset price risk with currency risk.

3.2 Budget Constraint

The investor enters period t with positions (n_{t-1}, A^k_{t-1}). After receiving all inflows at t, she chooses (C_t, n_t, A^k_t) and carries only the two risky legs (equity and crypto) into t+1. Thus, in LCU,

Inflows at t:

$$\underbrace{(P^s_t + D_t)\, n_{t-1}}_{\text{stock sale + dividend}} \quad + \quad \underbrace{\frac{S_t}{S_{t-1}}\,(1+\Theta^\star_t)\, A^k_{t-1}}_{\text{crypto proceeds in LCU after FX}} \quad + \quad \underbrace{I_t}_{\text{labor income}} \quad .$$

Outflows at t:

$$\underbrace{P^c_t C_t}_{\text{consumption}} \quad + \quad \underbrace{P^s_t n_t}_{\text{new stock position}} \quad + \quad \underbrace{A^k_t}_{\text{new crypto allocation}} \quad .$$

Hence the flow budget is

$$P^c_t C_t + P^s_t n_t + A^k_t \;=\; (P^s_t + D_t)n_{t-1} + \frac{S_t}{S_{t-1}}(1+\Theta^\star_t)A^k_{t-1} + I_t. \tag{1}$$

Define the LCU gross nominal and real returns on the crypto leg:

$$1 + R^k_{t+1} := \frac{S_{t+1}}{S_t}\,(1+\Theta^\star_{t+1}), \qquad 1 + R^{k,\text{real}}_{t+1} = \frac{1+R^k_{t+1}}{1+\pi_{t+1}}. \tag{2}$$

In logs (first-order), $r^k_{t+1} \approx \theta_{t+1} + \Delta s_{t+1}$.

Budget constraint. At the beginning of period t, the investor receives inflows from three sources. The stock position n_{t-1} yields dividends D_t and can be sold at the prevailing price P^s_t, giving

$$(P^s_t + D_t)n_{t-1}.$$

The crypto position A^k_{t-1} is measured in local currency. To compute its value at time t, we first convert it into USD using the exchange rate S_{t-1} (LCU per USD), obtaining $\frac{A^k_{t-1}}{S_{t-1}}$. This amount then grows with the USD crypto gross return $(1+\Theta^*_t)$ over period t, giving $(1+\Theta^*_t)\frac{A^k_{t-1}}{S_{t-1}}$. Finally, we convert back into local currency at the new exchange rate S_t, which yields

$$\frac{S_t}{S_{t-1}}(1+\Theta^*_t)\, A^k_{t-1}.$$

Finally, the investor receives labor income I_t. Thus total inflows are

$$(P^s_t + D_t)n_{t-1} \;+\; \frac{S_t}{S_{t-1}}(1+\Theta^*_t)A^k_{t-1} \;+\; I_t.$$

Outflows. During period t, resources are allocated to consumption $P^c_t C_t$, to stock purchases $P^s_t n_t$, and to crypto investment A^k_t. Thus total outflows are

$$P^c_t C_t + P^s_t n_t + A^k_t.$$

Flow budget. Equating inflows and outflows gives the period-t budget constraint:

$$P^c_t C_t + P^s_t n_t + A^k_t = (P^s_t + D_t)n_{t-1} + \frac{S_t}{S_{t-1}}(1+\Theta^*_t)A^k_{t-1} + I_t.$$

Crypto returns in local currency. The nominal gross return on crypto in local currency is

$$1 + R^k_{t+1} := \frac{S_{t+1}}{S_t}(1 + \Theta^*_{t+1}).$$

Adjusting for inflation, the real return is

$$1 + R^{k,\text{real}}_{t+1} = \frac{1 + R^k_{t+1}}{1 + \pi_{t+1}}.$$

A log-linear approximation further decomposes this into

$$r^k_{t+1} \approx \theta^*_{t+1} + \Delta s_{t+1},$$

where $\theta^*_{t+1} = \ln(1+\Theta^*_{t+1})$ is the log crypto return in USD, and $\Delta s_{t+1} = \ln\left(\frac{S_{t+1}}{S_t}\right)$ is the log foreign exchange return. This expression comes from writing

$$r^k_{t+1} = \ln\left(\tfrac{S_{t+1}}{S_t}(1 + \Theta^*_{t+1})\right) = \ln(1 + \Theta^*_{t+1}) + \ln\left(\tfrac{S_{t+1}}{S_t}\right).$$

3.3 Investor's Problem and SDF

Following standard arguments, we assume preferences are time-separable with discount factor $0 < \beta < 1$ and strictly increasing, concave $u(\cdot)$. The investor chooses $\{C_t, n_t, A^k_t\}_{t\geq 0}$ to

$$\max\ \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_t) \quad \text{s.t.} \quad (1), \quad n_t \geq 0,\ A^k_t \geq 0. \tag{3}$$

Define cash-on-hand (LCU) at time t:

$$M_t := (P^s_t + D_t)\, n_{t-1} \ + \ \frac{S_t}{S_{t-1}}(1 + \Theta^\star_t)\, A^k_{t-1} \ + \ I_t.$$

The investor chooses (C_t, n_t, A^k_t) to solve the Bellman problem

$$V(M_t) \ = \ \max_{C_t, n_t, A^k_t} \left\{ u(C_t) \ + \ \beta\, \mathbb{E}_t\big[V(M_{t+1})\big] \right\}$$

Investor's problem. The investor has time-separable preferences with discount factor $0 < \beta < 1$ and utility function $u(\cdot)$ that is strictly increasing and concave. The objective is to maximize expected discounted utility of consumption:

$$\max_{\{C_t, n_t, A^k_t\}_{t\geq 0}} \ \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_t),$$

subject to the period-by-period budget constraint, and with the non-negativity conditions $n_t \geq 0$ and $A^k_t \geq 0$.

Note: $\mathbb{E}_0[\cdot]$ means the expectation is taken **conditional on all information available at time** $t = 0$.

Cash-on-hand. Define cash-on-hand in local currency (LCU) at time t as

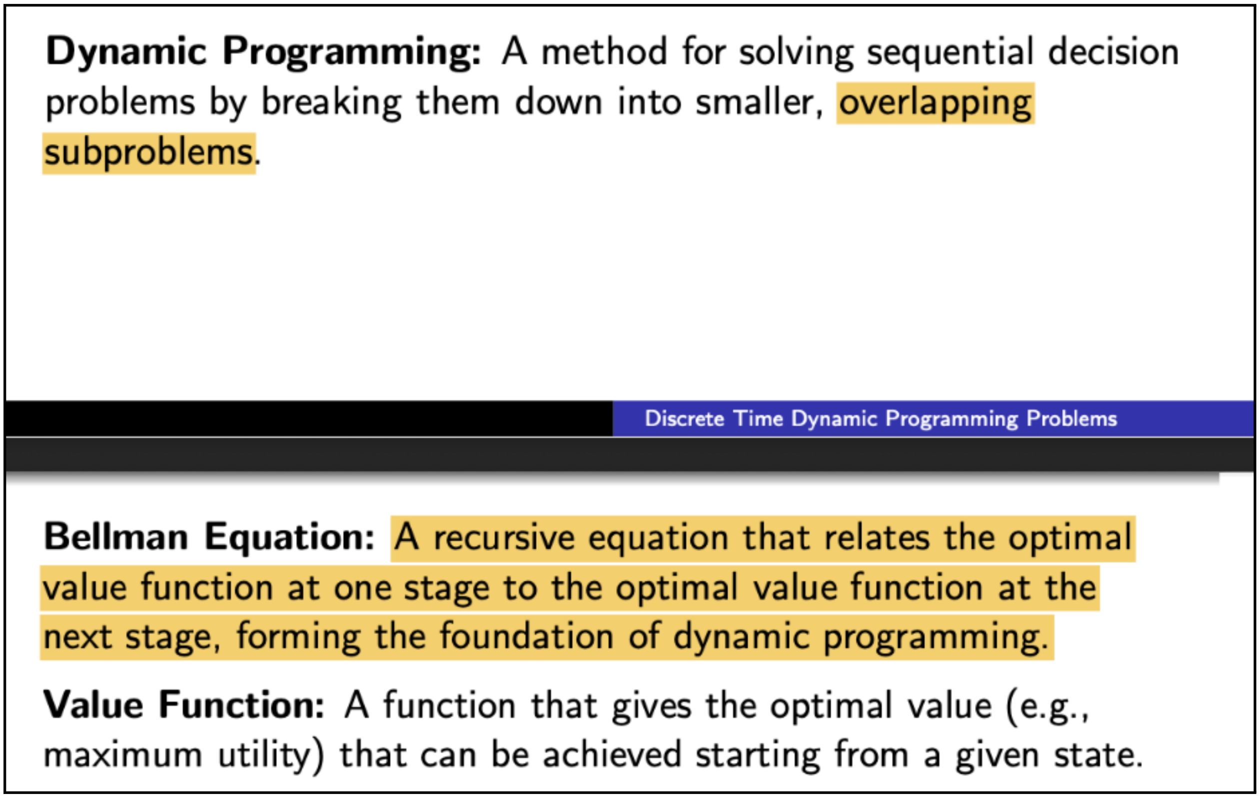
$$M_t := (P^s_t + D_t)n_{t-1} + \frac{S_t}{S_{t-1}}(1 + \Theta^*_t)A^k_{t-1} + I_t,$$

where the first term represents liquidating last period's stock holdings and receiving dividends, the second term is the crypto position converted into LCU after both the USD return and the FX movement, and the third term is exogenous labor income.

Bellman formulation. At time t, the investor chooses (C_t, n_t, A_t^k) to maximize lifetime utility. This leads to the Bellman equation

$$V(M_t) = \max_{C_t, n_t, A_t^k} \left\{ u(C_t) + \beta \, \mathbb{E}_t \big[V(M_{t+1}) \big] \right\}.$$

Here $V(M_t)$ is the value function, representing the maximum attainable expected utility given current resources M_t.



Value Function and Bellman Equation. The *value function* gives the maximum expected utility achievable from a given state. In our model, it represents the investor's discounted utility of consumption starting from cash-on-hand M_t.

The *Bellman equation* provides the recursive structure that underlies dynamic programming:

$$V(M_t) = \max_{C_t, n_t, A_t^k} \Big\{ u(C_t) + \beta \, \mathbb{E}_t \big[V(M_{t+1}) \big] \Big\}.$$

This relates today's value to tomorrow's, linking current utility with the expected continuation value.

At time zero, iterating this recursion forward recovers the original optimization problem:

$$V(M_0) = \max \, \mathbb{E}_0 \big[u(C_0) + \beta u(C_1) + \beta^2 u(C_2) + \cdots \big].$$

In practice, the Bellman equation is solved by *backward induction*: we start from the terminal period (or approximate one) and work backwards, using the recursive structure to determine optimal policies at each step.

subject to

$$P_t^c C_t \; + \; P_t^s n_t \; + \; A_t^k \; = \; M_t, \qquad M_{t+1} \; = \; (P_{t+1}^s + D_{t+1})\, n_t \; + \; \frac{S_{t+1}}{S_t}\,(1+\Theta_{t+1}^\star)\, A_t^k \; + \; I_{t+1}.$$

Let $V_M(\cdot)$ denote the derivative of V with respect to cash-on-hand. The standard first-order conditions lead to Euler equations in real LCU. Let the real SDF be

$$m_{t+1} := \beta\, \frac{u'(C_{t+1})}{u'(C_t)}\, \frac{1}{1+\pi_{t+1}}.$$

Then

$$\mathbb{E}_t\big[m_{t+1}\,(1+R_{t+1}^s)\big] = 1, \qquad \mathbb{E}_t\big[m_{t+1}\,(1+R_{t+1}^k)\big] = 1,$$

and, equivalently in real terms,

$$\mathbb{E}_t\big[m_{t+1}\,(1+R_{t+1}^{s,\mathrm{real}})\big] = 1, \qquad \mathbb{E}_t\big[m_{t+1}\,(1+R_{t+1}^{k,\mathrm{real}})\big] = 1.$$

Subtracting, we obtain the *risk-adjusted parity*:

$$\boxed{\mathbb{E}_t\Big[m_{t+1}\Big((1+R_{t+1}^s) - (1+R_{t+1}^k)\Big)\Big] = 0} \iff \mathbb{E}_t\big[m_{t+1}\,(R_{t+1}^{s,\mathrm{real}} - R_{t+1}^{k,\mathrm{real}})\big] = 0.$$

Let $w_{t+1} := R_{t+1}^{s,\mathrm{real}} - R_{t+1}^{k,\mathrm{real}}$. From the parity condition we have

$$E_t[w_{t+1}] = -\,\frac{Cov_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]}. \tag{2}$$

Decompose w_{t+1} around its conditional mean:

$$w_{t+1} = E_t[w_{t+1}] + \varepsilon_{t+1}, \qquad E_t[\varepsilon_{t+1}] = 0. \tag{3}$$

$$\varepsilon_{t+1} := w_{t+1} - E_t[w_{t+1}]. \tag{3'}$$

Since $R_{t+1}^{s,\mathrm{real}} = R_{t+1}^{k,\mathrm{real}} + w_{t+1}$, it follows that

$$R_{t+1}^{s,\mathrm{real}} = R_{t+1}^{k,\mathrm{real}} - \frac{Cov_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]} + \varepsilon_{t+1}, \qquad E_t[\varepsilon_{t+1}] = 0. \tag{4}$$

Deep Dive: Risk-Adjusted Parity and Wedge Decomposition

The Foundation: Why the Real SDF Takes This Form The real stochastic discount factor is

$$m_{t+1} := \beta \frac{u'(C_{t+1})}{u'(C_t)} \cdot \frac{1}{1+\pi_{t+1}}.$$

The logic is simple. Giving up one unit of consumption today has a marginal cost of $u'(C_t)$. The benefit of shifting that unit to tomorrow is $u'(C_{t+1})$, multiplied by the discount factor β and divided by $1+\pi_{t+1}$ to express payoffs in real terms. Economically, m_{t+1} is the marginal rate of substitution between consumption tomorrow and today. It reflects three elements: time preference (β), diminishing marginal utility ($u'(C_{t+1})/u'(C_t)$), and the inflation adjustment ($(1+\pi_{t+1})^{-1}$).

Clear Example: Is This Bond Worth Buying?

Suppose you face the choice between consuming \$50 today or investing it in a bond. Your current consumption is $C_t = 100$, and the bond costs \$50 and pays \$55 nominal next year, a 10% nominal return. Inflation is $\pi = 0.03$, so prices rise by 3%. Utility is logarithmic, $u(C) = \ln(C)$, with marginal utility $u'(C) = 1/C$. Your time preference is $\beta = 0.96$, reflecting mild impatience, and you expect future consumption income of $C_{t+1} = 105$ even without the bond.

Step 1: Cost today. Buying the bond reduces today's consumption from 100 to 50. The utility loss is given by

$$u'(100) \times 50 = \frac{1}{100} \times 50 = 0.50 \text{ utils.}$$

Step 2: Benefit tomorrow. Next year the bond pays \$55, but with 3% inflation the real payoff is $\frac{55}{1.03} = 53.40$. This increases tomorrow's consumption from 105 to 158.40. Tomorrow's marginal utility is $u'(105) = 1/105 = 0.00952$ utils per dollar, so the utility gain from the extra 53.40 is $0.00952 \times 53.40 = 0.508$ utils. Discounting back to today yields $\beta \times 0.508 = 0.96 \times 0.508 = 0.488$ utils.

Step 3: Decision. Comparing cost and benefit,

$$\underbrace{0.50}_{\text{loss today}} > \underbrace{0.488}_{\text{discounted gain}},$$

so you lose more than you gain. It is optimal to consume the \$50 today rather than buy the bond.

Step 4: SDF connection. The ratio of benefit to cost is

$$\frac{0.488}{0.50} = 0.976.$$

Writing this out,

$$\begin{aligned}
\frac{0.488}{0.50} &= \frac{\beta\, u'(105)\, 53.40}{u'(100)\, 50} \\
&= \underbrace{\beta \frac{u'(C_{t+1})}{u'(C_t)} \frac{1}{1+\pi}}_{m_{t+1}=0.887} \times \underbrace{\frac{53.40}{50}}_{1.068 \text{ real return}} \\
&= 0.887 \times 1.068 = 0.947.
\end{aligned}$$

Here $m_{t+1} = 0.887$ is the stochastic discount factor, which can be interpreted as

$$m_{t+1} = \frac{\text{Value of \$1 tomorrow in today's utility terms}}{\text{Value of \$1 today}}.$$

Since $m_{t+1} \times \text{real return} = 0.947 < 1$, each dollar invested in the bond gives back less than a dollar's worth of value. The optimal choice is to reject the bond.

The Euler Equations: No-Arbitrage Conditions

For any asset i with gross *real* return $(1 + R^i_{t+1})$, the optimality condition is

$$\mathbb{E}_t\left[m_{t+1}(1 + R^i_{t+1})\right] = 1.$$

Interpretation. This states that the expected payoff of investing one unit today, when discounted by the stochastic discount factor m_{t+1}, must equal its cost of one unit today. If this condition were violated, there would exist an arbitrage opportunity.

Derivation. Start from the marginal utility trade-off. If you give up one unit of consumption today, the cost in utility terms is

$$\text{Cost} = u'(C_t).$$

If you invest that unit, it grows into $(1 + R^i_{t+1})$ units tomorrow. The benefit in utility terms is

$$\text{Benefit} = \beta\, \mathbb{E}_t\left[u'(C_{t+1})(1 + R^i_{t+1})\right].$$

Why this is the marginal benefit. At time t, if you invest one unit in asset i, it grows into $(1 + R^i_{t+1})$ units next period. Multiplying this payoff by $u'(C_{t+1})$ gives the marginal utility impact $u'(C_{t+1})(1 + R^i_{t+1})$. Because both the payoff and $u'(C_{t+1})$ are uncertain at time t, we discount and take the conditional expectation, so the expected marginal benefit of investing one unit today is

$$\beta\,\mathbb{E}_t\big[u'(C_{t+1})(1 + R^i_{t+1})\big].$$

This is precisely the extra expected utility obtained by shifting one unit of consumption from today to tomorrow via asset i.

At the optimum, cost and benefit are equal:

$$u'(C_t) = \beta\,\mathbb{E}_t\big[u'(C_{t+1})(1 + R^i_{t+1})\big]\,.$$

Rearranging. Divide both sides by $u'(C_t)$:

$$1 = \mathbb{E}_t\left[\beta\frac{u'(C_{t+1})}{u'(C_t)}(1 + R^i_{t+1})\right].$$

Define the stochastic discount factor

$$m_{t+1} := \beta\frac{u'(C_{t+1})}{u'(C_t)} \cdot \frac{1}{1 + \pi_{t+1}},$$

where the last term deflates nominal payoffs into real terms. Substituting this definition yields the compact Euler equation:

$$\mathbb{E}_t\big[m_{t+1}(1 + R^i_{t+1})\big] = 1.$$

Key idea. The Euler equation enforces that, in equilibrium, every asset's expected *utility-adjusted* return is the same as the cost of one unit today. This is what rules out arbitrage in a consumption-based model. This must hold for *every* asset the investor holds.

Subtracting Euler Equations: The Parity Condition

For stocks: $\mathbb{E}_t[m_{t+1}(1 + R^{s,\text{real}}_{t+1})] = 1$

For crypto: $\mathbb{E}_t[m_{t+1}(1 + R^{k,\text{real}}_{t+1})] = 1$

Subtracting:

$$\begin{aligned}
\mathbb{E}_t[m_{t+1}(1 + R^{s,\text{real}}_{t+1})] - \mathbb{E}_t[m_{t+1}(1 + R^{k,\text{real}}_{t+1})] &= 0\\
\mathbb{E}_t\left[m_{t+1}\left((1 + R^{s,\text{real}}_{t+1}) - (1 + R^{k,\text{real}}_{t+1})\right)\right] &= 0\\
\mathbb{E}_t\left[m_{t+1}(R^{s,\text{real}}_{t+1} - R^{k,\text{real}}_{t+1})\right] &= 0
\end{aligned}$$

Why is this powerful? This is NOT saying expected returns are equal. It's saying the SDF-weighted expected return difference is zero. This is a restriction on the *joint distribution* of $(m_{t+1}, R^s_{t+1}, R^k_{t+1})$.

The Covariance Decomposition: Extracting Economic Content

Define the wedge: $w_{t+1} := R^{s,\text{real}}_{t+1} - R^{k,\text{real}}_{t+1}$

We have: $\mathbb{E}_t[m_{t+1}w_{t+1}] = 0$

Use the fundamental decomposition of expectations:

$$\mathbb{E}_t[XY] = \mathbb{E}_t[X] \cdot \mathbb{E}_t[Y] + \text{Cov}_t(X, Y)$$

Applied to our case:

$$\begin{aligned} 0 &= \mathbb{E}_t[m_{t+1}w_{t+1}] \\ &= \mathbb{E}_t[m_{t+1}] \cdot \mathbb{E}_t[w_{t+1}] + \mathrm{Cov}_t(m_{t+1}, w_{t+1}) \end{aligned}$$

Solving for the expected wedge:

$$\mathbb{E}_t[w_{t+1}] = -\frac{\mathrm{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]} \tag{2}$$

This is equation (2) - the KEY result.

Interpreting the Covariance Term

From the Euler equations we know that, for every asset i,

$$\mathbb{E}_t\left[m_{t+1}(1 + R^i_{t+1})\right] = 1.$$

Applying this to both stocks and crypto and subtracting, we obtained the *parity condition*

$$\mathbb{E}_t\left[m_{t+1}\left(R^s_{t+1} - R^k_{t+1}\right)\right] = 0.$$

Defining the return wedge as $w_{t+1} := R^s_{t+1} - R^k_{t+1}$, this condition becomes

$$\mathbb{E}_t[m_{t+1}w_{t+1}] = 0.$$

To interpret this, decompose the expectation into mean and covariance:

$$\mathbb{E}_t[m_{t+1}w_{t+1}] = \mathbb{E}_t[m_{t+1}]\,\mathbb{E}_t[w_{t+1}] + \mathrm{Cov}_t(m_{t+1}, w_{t+1}).$$

Setting this equal to zero and solving for the expected wedge gives

$$\mathbb{E}_t[w_{t+1}] = -\frac{\mathrm{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]}.$$

Interpretation. The condition

$$\mathbb{E}_t[w_{t+1}] = -\frac{\mathrm{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]}, \qquad w_{t+1} = R^s_{t+1} - R^k_{t+1},$$

What are we looking at?

- $w_{t+1} = R^s_{t+1} - R^k_{t+1}$ = how much better stocks do than crypto
- m_{t+1} = the SDF = high when times are BAD (recession, low consumption)
- The equation says: expected stock outperformance depends on whether stocks beat crypto in good times or bad times

Case 1. $\mathrm{Cov}_t(m_{t+1}, w_{t+1}) > 0$
Interpretation: When m_{t+1} goes up (bad times), w_{t+1} also goes up. Translation: stocks beat crypto more in recessions. For example, suppose a recession hits. Stocks fall 20%, but crypto falls 40%. Then

$$w_{t+1} = -20\% - (-40\%) = +20\%,$$

which is positive. This matters because you really need money in recessions, and stocks serve as better insurance. As a result, investors accept lower average returns on stocks—they are effectively paying for insurance. Hence,

$$\mathbb{E}_t[w_{t+1}] < 0,$$

which means crypto has higher expected returns on average.

Case 2. $\quad \text{Cov}_t(m_{t+1}, w_{t+1}) < 0$

Interpretation: When m_{t+1} goes up (bad times), w_{t+1} goes down. Translation: stocks beat crypto less in recessions, or may even lose to crypto.
For example, suppose a recession hits. Stocks fall 40%, but crypto only falls 20%. Then

$$w_{t+1} = -40\% - (-20\%) = -20\%,$$

which is negative. This matters because stocks fail you when you need them most, meaning they are bad insurance. As a result, investors demand higher average returns to hold stocks as compensation for this risk. Hence,

$$\mathbb{E}_t[w_{t+1}] > 0,$$

which means stocks have higher expected returns on average.

The Punchline: Whichever asset does relatively worse in recessions must pay higher average returns to get investors to hold it.

The Wedge Decomposition

Any random variable can be decomposed as:

$$w_{t+1} = \mathbb{E}_t[w_{t+1}] + \varepsilon_{t+1} \tag{3}$$

where $\mathbb{E}_t[\varepsilon_{t+1}] = 0$ by construction. This defines:

$$\varepsilon_{t+1} := w_{t+1} - \mathbb{E}_t[w_{t+1}]$$

Properties of ε_{t+1}:

- Mean zero: $\mathbb{E}_t[\varepsilon_{t+1}] = 0$
- Unpredictable given information at t
- Orthogonal to any time-t variables
- Represents *idiosyncratic* shocks

The Final Decomposition: Equation (4)

Starting from the definition: $w_{t+1} = R^{s,\text{real}}_{t+1} - R^{k,\text{real}}_{t+1}$

Rearranging: $R^{s,\text{real}}_{t+1} = R^{k,\text{real}}_{t+1} + w_{t+1}$

Substitute equations (2) and (3):

$$\begin{aligned} R^{s,\text{real}}_{t+1} &= R^{k,\text{real}}_{t+1} + \mathbb{E}_t[w_{t+1}] + \varepsilon_{t+1} \\ &= R^{k,\text{real}}_{t+1} - \frac{\text{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]} + \varepsilon_{t+1} \end{aligned}$$

$$\boxed{R^{s,\text{real}}_{t+1} = R^{k,\text{real}}_{t+1} - \frac{\text{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]} + \varepsilon_{t+1}} \tag{4}$$

Economic Interpretation of Equation (4)

This equation reveals the fundamental structure of asset returns.

Component 1: $R_{t+1}^{k,\mathbf{real}}$ **(Baseline return).** Here the cryptocurrency serves as the reference asset. The decomposition could equivalently be written the other way around, solving for crypto in terms of stocks instead of stocks in terms of crypto.

Component 2: $-\dfrac{\mathrm{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]}$ **(Systematic risk adjustment).** This part is predictable at time t and depends on the covariance structure with marginal utility. It represents the *risk premium* that ensures both assets are fairly priced. The negative sign is crucial: if stocks covary more positively with m_{t+1}—meaning they are better insurance—then they must earn lower expected returns.

Component 3: ε_{t+1} **(Idiosyncratic shock).** This term is unpredictable at time t, has mean zero, and is orthogonal to m_{t+1}. It captures asset-specific noise, which can be diversified away and therefore does not command a risk premium, since by construction $\mathbb{E}_t[\varepsilon_{t+1}] = 0$.

Corollary Under the model condition, projecting $R_{t+1}^{s,\text{real}}$ on $R_{t+1}^{k,\text{real}}$ yields

$$R_{t+1}^{s,\text{real}} = \alpha + \beta\, R_{t+1}^{k,\text{real}} + u_{t+1}, \qquad E_t[u_{t+1}] = 0,$$

where α captures the average mean adjustment, β measures the strength of comovement, and u_{t+1} is the mean-zero wedge innovation. The model predicts $\beta > 0$ and $\beta \to 1$ as the idiosyncratic wedge variance vanishes.

The structural parity condition implies

$$R_{t+1}^{s,\text{real}} = R_{t+1}^{k,\text{real}} - \frac{Cov_t(m_{t+1}, w_{t+1})}{E_t[m_{t+1}]} + \varepsilon_{t+1}, \qquad E_t[\varepsilon_{t+1}] = 0,$$

so the conditional mean of the wedge is determined by the interaction of the pricing kernel m_{t+1} with the wedge w_{t+1}. In the empirical regression

$$R_{t+1}^{s,\text{real}} = \alpha + \beta R_{t+1}^{k,\text{real}} + u_{t+1},$$

this adjustment cannot be observed directly. Its average effect is absorbed into the intercept α, while any time-varying deviations not captured by α appear in the residual u_{t+1}. Thus α and u_{t+1} together proxy for the SDF–wedge term in empirical implementation.

We now generate the main testable proposition linking the domestic equity and crypto assets. We now state two testable propositions that link the domestic equity leg and the FX-converted crypto leg implied by the first order condition.

Corollary: Regression Form and Structural Link

Projecting stock returns on crypto returns produces the regression

$$R_{t+1}^{s,\text{real}} = \alpha + \beta R_{t+1}^{k,\text{real}} + u_{t+1}, \qquad \mathbb{E}_t[u_{t+1}] = 0.$$

Interpretation of coefficients.

- α is the regression intercept. It captures the *average difference* between stock and crypto returns that remains even after accounting for comovement. In the model, this corresponds to the time–average of the wedge correction
$$-\frac{\mathrm{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]}.$$
- β is the slope coefficient. It measures the strength of comovement between stock and crypto returns. If $\beta = 1$, the two assets move one–for–one and only the wedge correction matters. If $\beta < 1$, stocks

respond less to crypto movements; if $\beta > 1$, they respond more. The model predicts $\beta > 0$ and $\beta \to 1$ as idiosyncratic wedge variance disappears.

- u_{t+1} is the residual. It absorbs *all deviations not explained by α and $\beta R^{k,real}_{t+1}$*. Formally, it includes both short–run fluctuations in the wedge adjustment and the mean–zero innovation ε_{t+1} from the decomposition:

$$u_{t+1} \approx \left(- \frac{\mathrm{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]} - \alpha \right) + \varepsilon_{t+1}.$$

Structural comparison. The wedge decomposition gives the theoretical relation

$$R^{s,\text{real}}_{t+1} = R^{k,\text{real}}_{t+1} - \frac{\mathrm{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]} + \varepsilon_{t+1}, \qquad \mathbb{E}_t[\varepsilon_{t+1}] = 0.$$

Lining this up with the regression form shows:

$$R^{s,\text{real}}_{t+1} = \underbrace{R^{k,\text{real}}_{t+1}}_{\text{explained by } \beta R^{k,\text{real}}_{t+1}} + \underbrace{\left(- \frac{\mathrm{Cov}_t(m_{t+1}, w_{t+1})}{\mathbb{E}_t[m_{t+1}]} \right)}_{\text{absorbed into } \alpha} + \underbrace{\varepsilon_{t+1}}_{\text{residual part of } u_{t+1}}.$$

Summary. The regression is simply the projection of one return on another. Each component has a direct structural counterpart:

$$\alpha \approx \text{average wedge correction}, \qquad \beta \approx \text{strength of comovement}, \qquad u_{t+1} \approx \text{time–varying correction + noise}.$$

Thus the empirical regression is not just statistical: it is a re–expression of the theoretical wedge relation in estimable form.

3.4 Main Proposition

In the short run, after controlling for common shocks,

$$r^{s,\text{real}}_{t+1} = \alpha + \beta\, r^{k,\text{real}}_{t+1} + u_{t+1}.$$

Let ε^s_{t+1} and ε^k_{t+1} be the idiosyncratic components of $r^{s,\text{real}}_{t+1}$ and $r^{k,\text{real}}_{t+1}$ after removing common shocks, and define the idiosyncratic parity wedge $e_{t+1} := \varepsilon^s_{t+1} - \varepsilon^k_{t+1}$. Then

$$\beta = \frac{\mathrm{Cov}(\varepsilon^s_{t+1}, \varepsilon^k_{t+1})}{\mathrm{Var}(\varepsilon^k_{t+1})} = 1 + \frac{\mathrm{Cov}(e_{t+1}, \varepsilon^k_{t+1})}{\mathrm{Var}(\varepsilon^k_{t+1})},$$

and by Cauchy–Schwarz,

$$1 - \sqrt{\frac{\mathrm{Var}(e_{t+1})}{\mathrm{Var}(\varepsilon^k_{t+1})}} \le \beta \le 1 + \sqrt{\frac{\mathrm{Var}(e_{t+1})}{\mathrm{Var}(\varepsilon^k_{t+1})}}.$$

In particular, if $\mathrm{Var}(e_{t+1})/\mathrm{Var}(\varepsilon^k_{t+1}) < 1$, then $\beta > 0$; moreover, as $\mathrm{Var}(e_{t+1})/\mathrm{Var}(\varepsilon^k_{t+1}) \to 0$, we have $\beta \to 1$ as required. [1]

Detailed Explanation. We start from the short–run regression of real stock returns on real crypto returns:

$$r^{s,\text{real}}_{t+1} = \alpha + \beta r^{k,\text{real}}_{t+1} + u_{t+1}.$$

Here β is the slope coefficient of interest. It measures how much stocks move when crypto moves, after controlling for common shocks. To understand the determinants of β, we strip out common shocks and focus on the idiosyncratic components.

Define ε^s_{t+1} and ε^k_{t+1} to be the *idiosyncratic parts* of $r^{s,\text{real}}_{t+1}$ and $r^{k,\text{real}}_{t+1}$ respectively. Intuitively, these are the movements unique to each asset once we remove what they both share in common.

Next, define the *idiosyncratic wedge*

$$e_{t+1} := \varepsilon^s_{t+1} - \varepsilon^k_{t+1}.$$

This wedge measures how much stock–specific noise differs from crypto–specific noise.

By properties of linear regression, the slope β can be written as the covariance of the dependent variable with the regressor divided by the variance of the regressor:

$$\beta = \frac{\operatorname{Cov}(\varepsilon^s_{t+1}, \varepsilon^k_{t+1})}{\operatorname{Var}(\varepsilon^k_{t+1})}.$$

This formula simply says: the slope tells us how much the two idiosyncratic parts move together, scaled by how volatile crypto's idiosyncratic component is.

Now expand ε^s_{t+1} using the wedge definition:

$$\varepsilon^s_{t+1} = e_{t+1} + \varepsilon^k_{t+1}.$$

Substituting into the covariance gives

$$\operatorname{Cov}(\varepsilon^s_{t+1}, \varepsilon^k_{t+1}) = \operatorname{Cov}(e_{t+1} + \varepsilon^k_{t+1}, \varepsilon^k_{t+1}) = \operatorname{Cov}(e_{t+1}, \varepsilon^k_{t+1}) + \operatorname{Var}(\varepsilon^k_{t+1}).$$

Dividing by $\operatorname{Var}(\varepsilon^k_{t+1})$ yields

$$\beta = 1 + \frac{\operatorname{Cov}(e_{t+1}, \varepsilon^k_{t+1})}{\operatorname{Var}(\varepsilon^k_{t+1})}.$$

So β is equal to one, plus a correction term coming from the covariance between the wedge and crypto's idiosyncratic return.

To bound this correction term, apply the Cauchy–Schwarz inequality:

$$|\operatorname{Cov}(e_{t+1}, \varepsilon^k_{t+1})| \le \sqrt{\operatorname{Var}(e_{t+1}) \cdot \operatorname{Var}(\varepsilon^k_{t+1})}.$$

Dividing through by $\operatorname{Var}(\varepsilon^k_{t+1})$ gives

$$-\sqrt{\frac{\operatorname{Var}(e_{t+1})}{\operatorname{Var}(\varepsilon^k_{t+1})}} \;\le\; \frac{\operatorname{Cov}(e_{t+1}, \varepsilon^k_{t+1})}{\operatorname{Var}(\varepsilon^k_{t+1})} \;\le\; \sqrt{\frac{\operatorname{Var}(e_{t+1})}{\operatorname{Var}(\varepsilon^k_{t+1})}}.$$

Adding 1 to all sides produces the inequality

$$1 - \sqrt{\frac{\operatorname{Var}(e_{t+1})}{\operatorname{Var}(\varepsilon^k_{t+1})}} \;\le\; \beta \;\le\; 1 + \sqrt{\frac{\operatorname{Var}(e_{t+1})}{\operatorname{Var}(\varepsilon^k_{t+1})}}.$$

This inequality places β in a narrow interval around 1. To interpret:

- When the wedge variance $\operatorname{Var}(e_{t+1})$ is small compared to crypto's idiosyncratic variance $\operatorname{Var}(\varepsilon^k_{t+1})$, the square–root term is small. This means β cannot stray far from 1.
- If the ratio

  $$\frac{\operatorname{Var}(e_{t+1})}{\operatorname{Var}(\varepsilon^k_{t+1})} < 1,$$

  then even the lower bound of the interval is positive, which guarantees $\beta > 0$.

- In the extreme case where
$$\frac{\mathrm{Var}(e_{t+1})}{\mathrm{Var}(\varepsilon^k_{t+1})} \to 0,$$
the square–root term vanishes entirely, so the bounds collapse to $\beta = 1$ exactly.

In plain language: β measures how closely stock and crypto idiosyncratic returns move together. If the wedge noise is large, β may depart from 1. But as wedge noise shrinks, the two assets' idiosyncratic parts align more and more tightly, forcing β to converge to 1.